Analysis of Smartphone Usage and Work Productivity using KNIME

Course: Machine and Deep Learning for Business.

Introduction:

Smartphones are widely used in everyday life for communication, work, and entertainment. However, excessive smartphone usage can affect productivity and daily performance. Understanding the relationship between smartphone usage behaviour and productivity can help identify patterns that influence work efficiency. In this project, data analysis and machine learning techniques were applied using the KNIME Analytics Platform to analyze how smartphone usage habits affect work productivity. The analysis includes data preprocessing, descriptive statistics, data visualization, and predictive modelling to explore patterns between smartphone usage behaviour and productivity levels.

Dataset Description:

The dataset used in this analysis contains information about smartphone usage behaviour and productivity levels. It includes variables related to user demographics, phone usage patterns, sleep habits, and stress levels. The dataset contains approximately **50,000 observations and multiple variables**, including both categorical and numerical attributes. The most important variable in this analysis is **Work Productivity Score**, which represents the productivity level of users.

Dataset Variables Table:

Variable	Type	Description
Age	Numeric	Age of the user
Gender	Categorical	Male or Female
Occupation	Categorical	User occupation
Device_Type	Categorical	Android or iOS device
Daily_Phone_Hours	Numeric	Average phone usage per day
Social_Media_Hours	Numeric	Time spent on social media
Sleep_Hours	Numeric	Average sleep hours per day
Stress_Level	Numeric	Reported stress level
App_Usage_Count	Numeric	Number of apps used
Caffeine_Intake_Cups	Numeric	Cups of caffeine consumed daily
Weekend_Screen_Time_Hours	Numeric	Phone usage during weekends
Work_Productivity_Score	Numeric	Productivity score

Research Question:

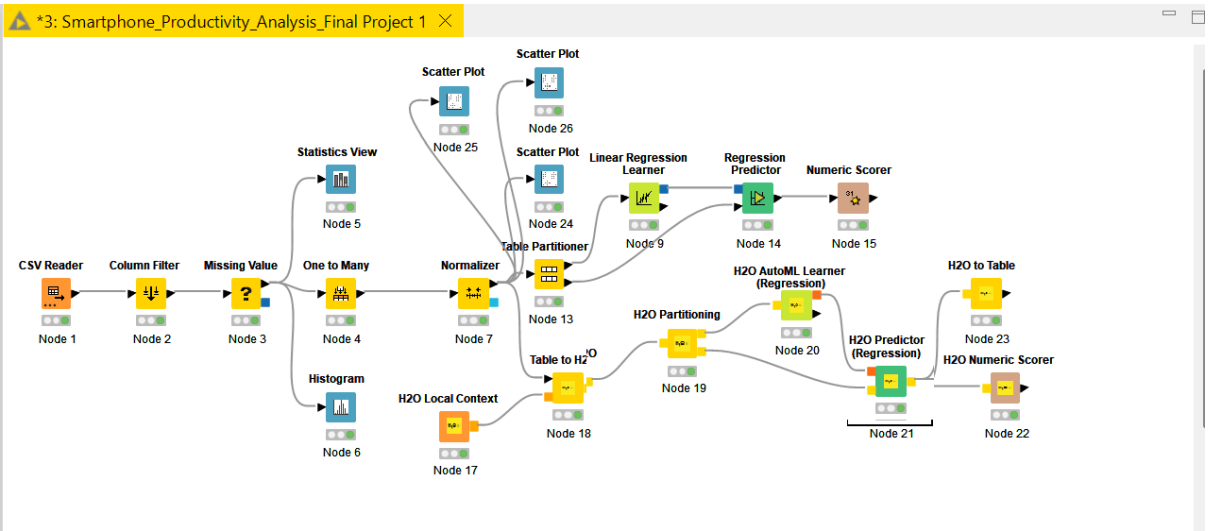
Q. How do smartphone usage patterns influence work productivity levels?

A. This analysis aims to explore the relationship between smartphone usage behavior and productivity by examining variables such as daily phone usage, social media activity, sleep hours,

stress levels, and other behavioral factors. Machine learning models are used to predict productivity scores based on these variables and evaluate their predictive performance.

KNIME Workflow:

The data analysis was implemented using a KNIME workflow that includes data preprocessing, statistical analysis, visualization, and predictive modeling. The workflow consists of several steps. The dataset is first imported into KNIME and cleaned using preprocessing nodes. Categorical variables are converted into numerical format, and numerical variables are normalized to improve model performance. The dataset is then divided into training and testing sets using a table partitioning approach. Two predictive models were applied to analyze the dataset: Linear Regression and H2O AutoML Regression.



Descriptive Statistics:

Descriptive statistics were used to summarize the dataset and understand the distribution of key variables. The Statistics node in KNIME was used to calculate statistical measures such as mean, minimum, maximum, and standard deviation for each numeric variable. These statistics help provide an overview of the dataset and identify patterns in smartphone usage behavior and productivity scores.

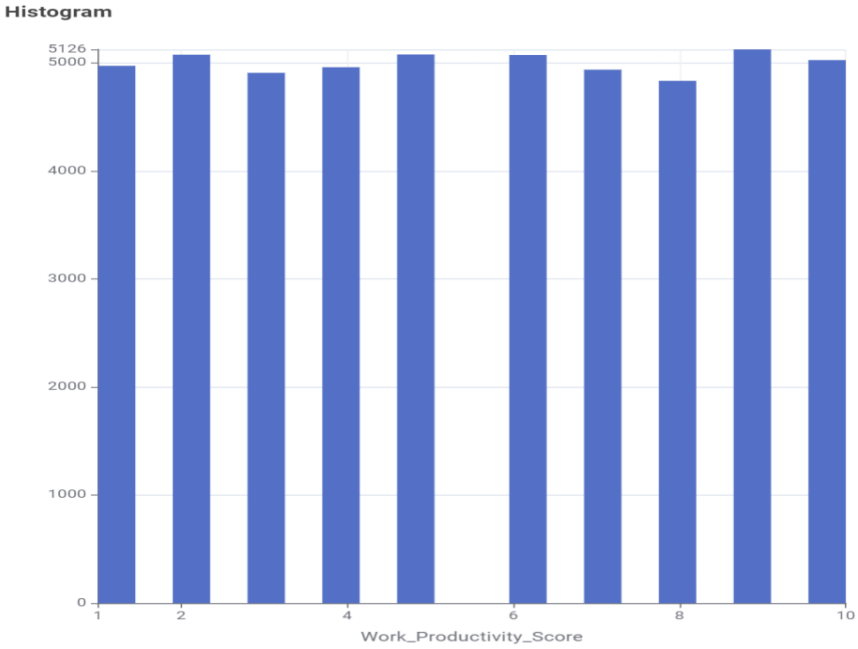
Statistics

Rows: 12 | Columns: 14

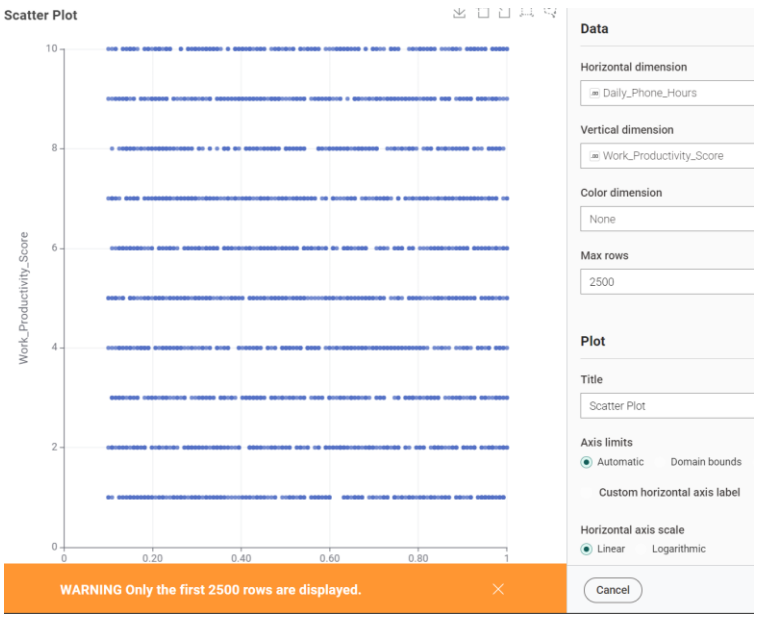
Name	Type	# Missing val...	# Unique val...	Minimum	Maximum	25%
Age	Number (Float)	0	43	18	60	28
Gender	String	0	3	?	?	?
Occupation	String	0	4	?	?	?
Device_Type	String	0	2	?	?	?
Daily_Phone_H	Number (Float)	0	111	1	12	3.8
Social_Media_H	Number (Float)	0	76	0.5	8	2.4
Work_Productiv	Number (Float)	0	10	1	10	3
Sleep_Hours	Number (Float)	0	51	4	9	5.2
Stress_Level	Number (Float)	0	10	1	10	3
App_Usage_Co	Number (Float)	0	56	5	60	18
Caffeine_Intake	Number (Float)	0	7	0	6	1
Weekend_Scree	Number (Float)	0	121	2	14	5

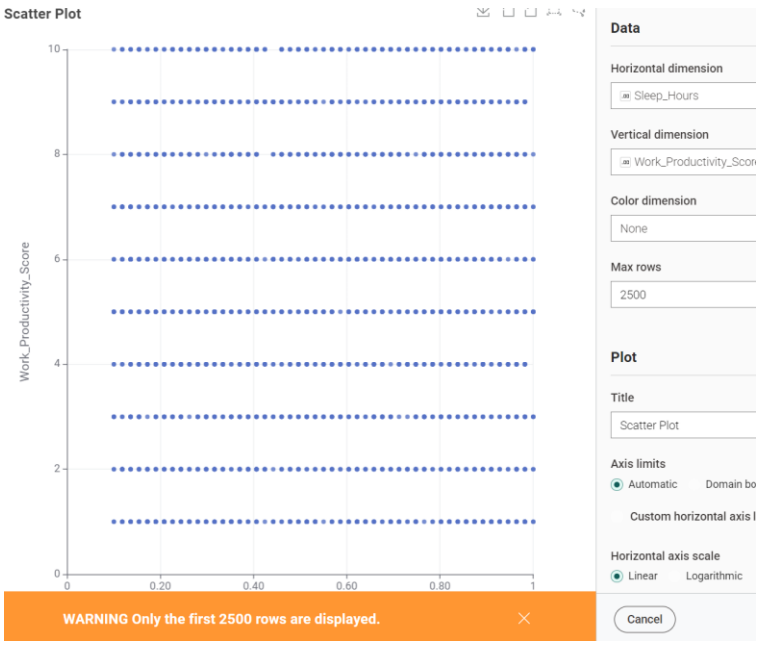
Data Visualization:

A histogram was created to visualize the distribution of the Work Productivity Score. Data visualization helps in understanding how productivity scores are distributed among users. The histogram provides a clear representation of how frequently different productivity scores occur within the dataset. Visualization also helps identify patterns or irregularities in the data.



In addition to the histogram, scatter plots were created to analyze the relationship between smartphone usage variables and work productivity score. Scatter plots were generated for variables such as **Daily Phone Hours, Social Media Hours, and Sleep Hours** against the Work Productivity Score to visually examine possible correlations.





Machine Learning Models:

Two machine learning models were applied to predict the Work Productivity Score. The first model is **Linear Regression**, which estimates the relationship between independent variables and the target variable. This model is commonly used for predicting continuous numerical values. The second model is **H2O AutoML**, which automatically tests multiple machine learning algorithms and selects the best performing model based on prediction accuracy. Applying multiple models allows comparison of model performance and provides a better understanding of prediction accuracy. The dataset was divided into training and testing sets using the Table Partitioner node in KNIME before applying the predictive models.

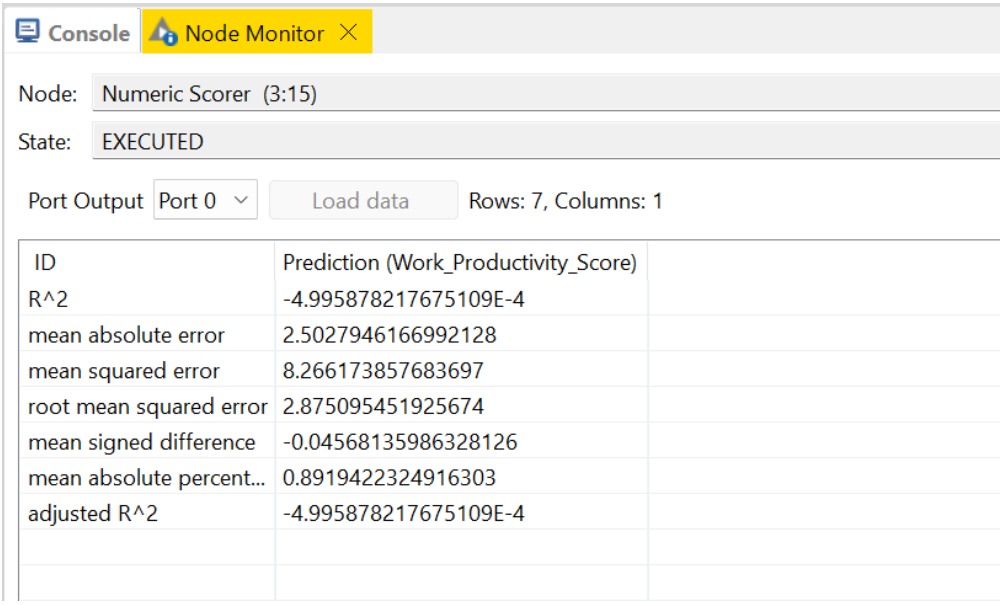
Model Results:

The performance of the models was evaluated using the Numeric Scorer node in KNIME. Evaluation metrics such as RMSE and MAE were used to measure prediction accuracy. Lower values of these metrics indicate better model performance.

Model	RMSE	R^2
Linear Regression	2.875	-0.00049
H2O AutoML	2.878	-0.00080

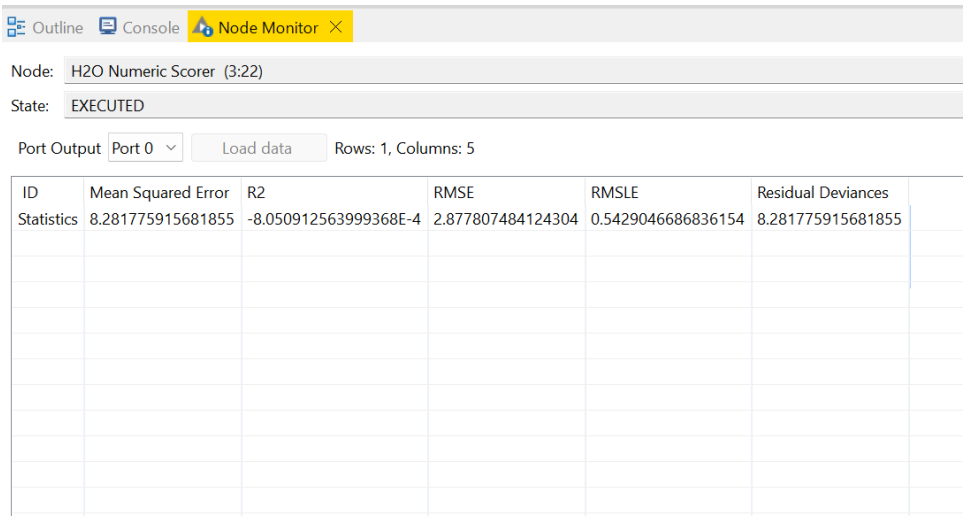
The performance of the models was evaluated using prediction error metrics such as Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). The Linear Regression model achieved an RMSE value of approximately **2.875**, while the H2O AutoML model achieved an RMSE value of approximately **2.878**. Since RMSE measures prediction error, lower values indicate better performance. In this analysis, both models produced very similar RMSE values. The R^2 values for both models are close to zero, indicating that the independent variables explain only a small portion of the variance in productivity scores. This suggests that smartphone usage variables alone may not fully explain work productivity levels.

Performance metrics of the Linear Regression model.



ID	Prediction (Work_Productivity_Score)
R^2	-4.995878217675109E-4
mean absolute error	2.5027946166992128
mean squared error	8.266173857683697
root mean squared error	2.875095451925674
mean signed difference	-0.04568135986328126
mean absolute percent...	0.8919422324916303
adjusted R^2	-4.995878217675109E-4

Performance metrics of the H2O AutoML regression model.



ID	Mean Squared Error	R2	RMSE	RMSLE	Residual Deviances
Statistics	8.281775915681855	-8.050912563999368E-4	2.877807484124304	0.5429046686836154	8.281775915681855

The performance of the models was evaluated using prediction error metrics such as RMSE and R^2. The RMSE values for both models are similar, indicating comparable prediction accuracy. The R^2 values are close to zero, which suggests that the models explain only a small portion of the variance in productivity scores. This indicates that smartphone usage variables alone may not fully explain productivity levels.

Discussion:

The analysis explored the relationship between smartphone usage behavior and work productivity. Descriptive statistics provided an overview of the dataset, while visualizations such as histograms and scatter plots helped observe patterns between variables like daily phone usage, social media hours, sleep hours, and the Work Productivity Score. The scatter plots show that productivity values appear in horizontal clusters because the score ranges from 1 to 10. Two machine learning models, Linear Regression and H2O AutoML, were applied to predict productivity levels. Both models produced similar results with RMSE values around **2.87**, while the **R^2 values were close to zero**, indicating that smartphone usage variables alone do not fully explain productivity levels.

Conclusion:

This project analyzed the relationship between smartphone usage behavior and work productivity using the KNIME Analytics Platform. The dataset was processed through several stages including data preprocessing, statistical analysis, visualization, and predictive modeling. Descriptive statistics and visualizations such as histograms and scatter plots helped identify patterns between smartphone usage variables and productivity levels. Two predictive models, Linear Regression and H2O AutoML, were applied to estimate the Work Productivity Score based on behavioral variables. The evaluation results showed that both models produced similar prediction accuracy with RMSE values around **2.87** and R^2 values close to zero, indicating that the independent variables explain only a limited portion of productivity variation. This suggests that productivity may also depend on additional factors beyond smartphone usage. Overall, the project demonstrates how data analytics and machine learning techniques can be used to explore behavioral patterns and analyze productivity-related data.

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